



POLITECNICO
MILANO 1863

Asset Allocation Module

Group 10

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Contents

0	Introduction	1
0.1	Dataset	1
0.2	Analysis Periods	1
1	Constrained and Robust Efficient Frontier	1
1.1	Methodology and Constraints	1
1.2	Results Analysis	2
2	Black–Litterman Model	3
2.1	Equilibrium Returns	3
2.2	Views Implementation	3
2.3	Posterior Expected Returns and Efficient Frontier	3
2.4	Portfolios under the BL model.	4
2.5	Contribution of Views and Portfolio Adjustments	5
2.6	Discussion of Effects	6
3	Diversification-Based Optimization	6
3.1	Methodology and Constraints	6
3.2	Results and Analysis	7
4	PCA and Conditional Value-at-Risk	7
4.1	Principal Component Analysis (PCA)	7
4.2	Portfolio I (PCA Maximum Sharpe).	7
4.3	Portfolio J (Minimum CVaR).	8
4.4	Comparative Results	8
5	Exercise 5: Personal Strategy	9
5.1	Rationale and Quantitative Intuition	9
5.2	Performance Evaluation Over Jan 2023 – Nov 2024	9
5.3	Discussion of Performance Indicators	9
5.4	Monthly Performance Results	10
5.5	Monthly Comparison of Performance	11
6	Weights Summary	11
7	Final Discussion: Out-of-Sample Analysis (2023–2024)	12
7.1	Out-of-Sample Performance of All Portfolios	12
7.2	Risk–Return Analysis	12
7.3	Out-of-Sample Performance Visualizations	14
7.4	Impact of macro-sensitivities on Portfolio behaviour	14

0 Introduction

This report summarizes the work carried out for the Asset Allocation Module (Exam C). The aim of the project is to build investment portfolios under specific models or constraints, compare their characteristics and evaluate the stability of results across different approaches and time periods.

0.1 Dataset

The study relies on a dataset of daily asset prices contained in the file `asset_prices.csv`, covering the period from October 10, 2001, to November 29, 2024. The dataset includes 16 anonymous indices belonging to the equity market.

Each index is assigned a macro-sensitivity label based on the information provided in `mapping_table.csv`:

- **Cyclical**
- **Neutral**
- **Defensive**

Furthermore, market-capitalization weights required for the Black-Litterman analysis are available in the file `capitalization_weights.csv`.

0.2 Analysis Periods

As specified in the project instructions, the analysis is divided into two main periods:

- **In-sample (calibration)**: January 2018 to December 2022, used to estimate parameters and construct the portfolios;
- **Out-of-sample (assessment)**: January 2023 to December 2024, employed to test the stability and effectiveness of the strategies on unseen data.

1 Constrained and Robust Efficient Frontier

This section discusses how the Mean-Variance efficient frontier is built when specific investment restrictions are introduced, and investigates how resampling procedures influence the stability and reliability of the resulting portfolios.

1.1 Methodology and Constraints

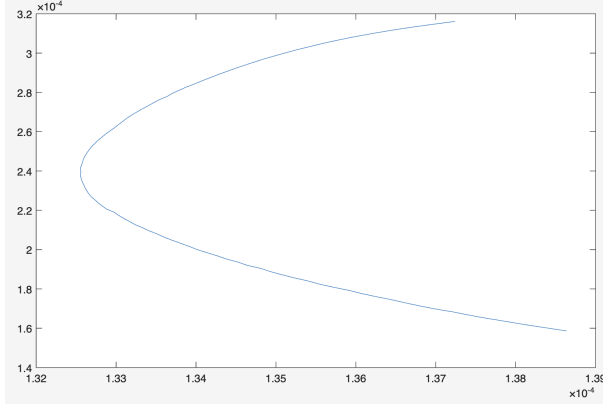
The portfolio optimization was carried out using the in-sample period from January 2018 to December 2022. Within the traditional Mean-Variance framework, we computed both the Minimum Variance Portfolio (MVP) and the Maximum Sharpe Ratio Portfolio (MSRP).

The asset weights w_i were required to satisfy the following conditions:

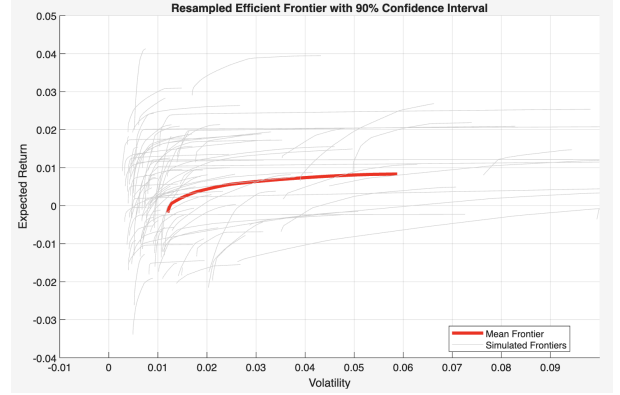
- **Full investment**: $\sum_i w_i = 1$.
- **No short selling**: $w_i \geq 0$ for every asset.
- **Maximum weight per asset**: individual asset weights limited to $w_i \leq 0.30$.
- **Macroeconomic assets**:

- Exposure to defensive assets not exceeding 45%;
- Exposure to neutral assets of at least 20%.

After defining the constrained frontier, we applied a resampling procedure, specifically, a Monte Carlo bootstrap was performed on the return dataset to generate 100 simulated versions of the efficient frontier. The robust portfolios (C and D) were then obtained by averaging the optimal portfolio weights across all simulations and selecting the minimum variance and the maximum Sharpe ones. This approach helps reduce the issue of “error maximization”, a well-known drawback of classical Mean–Variance optimization, where the algorithm tends to assets that appear artificially attractive due to random sampling noise.



(a) Efficient frontier of the nominal portfolio



(b) Resampled Efficient Frontier with 90% CI

1.2 Results Analysis

In our analysis we noticed that using only the built-in function for the frontier tends to highly concentrate the assets as a result of the high correlation, so we decided to use a manual minimization to build the frontier. This problem is avoided in the construction of the robust case due to the sampling procedure.

We can notice how the Robust estimations have worse results regarding the expected returns, highlighting the trade-off between the nominal and the robust approach.

The portfolios, both minimum variance and maximum Sharpe, present a big exposure to asset 4 and asset 5, while the biggest difference are in asset 12,13 and 15. This is explainable since these assets exhibit the lowest variances under $0.2 \cdot 10^{-3}$.

Table 1: In-sample performance metrics (2018–2022).

Portfolio	Exp. Return	Volatility	Sharpe Ratio
Portfolio A (Nominal MVP)	0.0608	0.18276	0.33277
Portfolio B (Nominal MSRP)	0.0797	0.18597	0.42837
Portfolio C (Robust MVP)	0.0556	0.19428	0.28624
Portfolio D (Robust MSRP)	0.0568	0.20258	0.28058

2 Black–Litterman Model

This section presents the implementation of the Black–Litterman (BL) framework, which blends equilibrium (market-implied) returns with subjective investor views. The process unfolds in three steps:

- computation of equilibrium returns,
- specification of investor views
- construction of posterior expected returns and efficient frontiers.

2.1 Equilibrium Returns

The equilibrium prior is derived from the market capitalization weights and a fixed risk–aversion parameter $\lambda = 1.2$. The implied expected return vector is computed as:

$$\Pi = \lambda \Sigma w_{\text{mkt}},$$

where Σ is the market covariance matrix and w_{mkt} the market portfolio weights.

A prior distribution of returns is visualized by drawing samples from $\mathcal{N}(\mu_{\text{mkt}}, C)$.

2.2 Views Implementation

Investor views are introduced through a pick matrix P , a vector of expected relative performances q , and a diagonal uncertainty matrix Ω . Each view has the general form

$$P \mathbb{E}[R] = q + \varepsilon.$$

In MATLAB, the objects (P, q, Ω) are produced by a custom function that assigns assets to macro-groups and translates the analyst’s beliefs into structured numerical restrictions.

2.3 Posterior Expected Returns and Efficient Frontier

The Black–Litterman posterior returns combine equilibrium and subjective components:

$$\mu_{\text{BL}} = (C^{-1} + P^{\top} \Omega^{-1} P)^{-1} (P^{\top} \Omega^{-1} q + C^{-1} \mu_{\text{mkt}}),$$

with posterior covariance

$$\Sigma_{\text{BL}} = (P^{\top} \Omega^{-1} P + C^{-1})^{-1}.$$

The following table reports the annualized prior and BL posterior expected returns produced by MATLAB.

Table 2: Annualized Prior vs. Black–Litterman Expected Returns

Asset	PriorReturnAnnual	BLReturnAnnual
Asset1	0.063762	0.052131
Asset2	0.040688	0.048091
Asset3	0.055293	0.038764
Asset4	0.036177	0.037800
Asset5	0.042927	0.035822
Asset6	0.061359	0.039662
Asset7	0.050481	0.021545
Asset8	0.058694	0.045009
Asset9	0.067869	0.091090
Asset10	0.056099	0.049402
Asset11	0.049744	0.044664
Asset12	0.053433	0.040942
Asset13	0.051777	0.039650
Asset14	0.053856	0.047135
Asset15	0.043261	0.038434
Asset16	0.057187	0.043176

Two efficient frontiers are then generated: the equilibrium frontier and the BL frontier. The BL frontier shifts upward, indicating that incorporating views enhances the feasible risk–return trade-off.

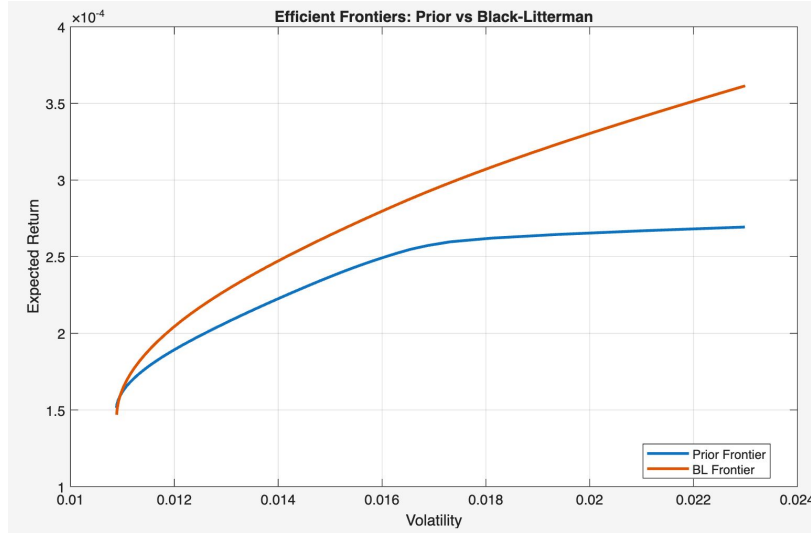


Figure 2: Efficient frontiers: Prior vs. Black–Litterman.

2.4 Portfolios under the BL model.

Using the posterior returns μ_{BL} and the BL covariance matrix Σ_{BL} , we optimize two benchmark portfolios: (i) the *BL Minimum Variance Portfolio* (BL-MinVar), and (ii) the *BL Maximum Sharpe Ratio Portfolio* (BL-MaxSharpe).

The BL-MinVar portfolio is highly concentrated in Asset4 and Asset5, with a small residual allocation to Asset8, while all other weights are effectively zero. In contrast, the BL-MaxSharpe portfolio spreads capital mainly across Asset2, Asset4 and Asset8, with a small position in Asset1, reflecting the views’ tilt towards these assets.

2.5 Contribution of Views and Portfolio Adjustments

The effect of each view on posterior returns is isolated by computing:

$$\Delta_i = \Sigma P_i (P_i^\top \Sigma P_i + \Omega_i)^{-1} (q_i - P_i^\top \mu_{\text{mkt}}),$$

which shows how each view shifts expected returns asset-by-asset.

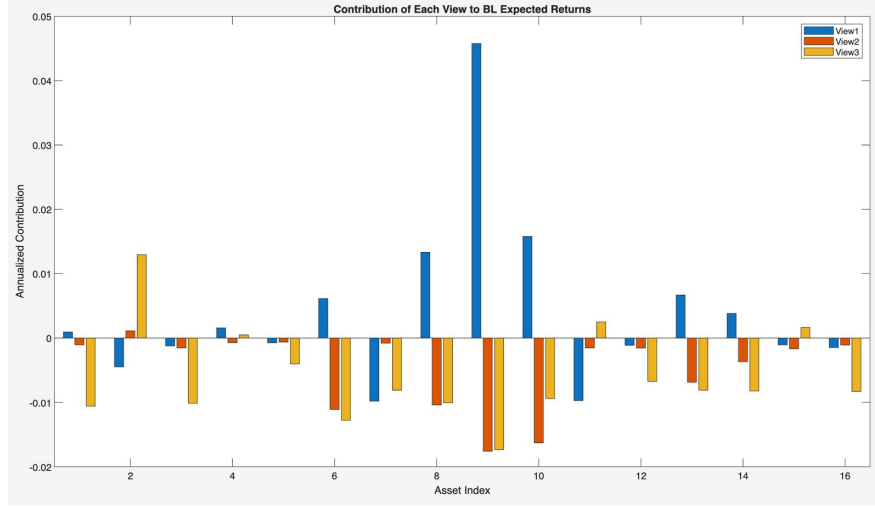


Figure 3: Contribution of each view to posterior expected returns.

Similarly, the changes in portfolio weights between the BL Maximum Sharpe portfolio and the market benchmark are displayed below.

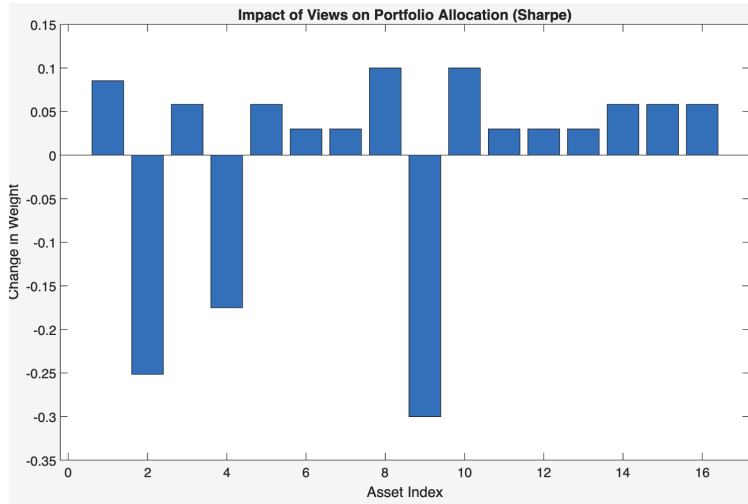


Figure 4: Impact of views on asset weights BL MSRP vs Prior MSRP

These results show how the incorporation of investor beliefs redistributes risk exposures and adjusts expected returns while preserving consistency with the overall market structure.

2.6 Discussion of Effects

The subjective views introduced into the Black-Litterman framework substantially reshaped the portfolio allocation relative to the market-equilibrium starting point, as shown in Figure 3 and Figure 4.

- **View 1:** As anticipated, the return contributions of cyclical assets (8, 9, 10) increase by 0.013, 0.016, and 0.046, respectively. Contrary to expectations, the allocation to the neutral Asset 6 also rises by 0.006, indicating that it nonetheless provides a meaningful contribution to the Sharpe in spite of the adverse view.
- **View 2:** Consistent with the forecast, the model sharply reduces the contribution of Asset 10 by 0.016, pushed by the expected slowdown compared to the Defensive assets (the most conservative ones).
- **View 3:** The allocation shifts align with the expressed view, the contribution of Asset 2 increases by 0.012, while that of Asset 13 decreases by 0.008.

3 Diversification-Based Optimization

The goal is to construct the Maximum Diversification Ratio Portfolio and the Maximum Entropy Portfolio in Risk Contribution in order to analyze in different ways the diversification of the portfolios.

3.1 Methodology and Constraints

To compute the Diversification Ratio for each contributor we use the following formula:

$$DR_i = \frac{w_i \sigma_i}{w \cdot \Sigma \cdot w}$$

where denominator is the volatility of the whole portfolio.

To compute the Entropy for each risk contributor we use the following formulas

$$\theta_i = w_i \frac{(\Sigma \cdot w)_i}{w \cdot \Sigma \cdot w}$$

$$H = - \sum_{i=1}^N \theta_i \ln(\theta_i)$$

The asset weights w_i were required to satisfy the following conditions:

- Full investment: $\sum_i w_i = 1$.
- No short selling: $w_i \geq 0$ for every asset.
- Maximum weight per asset: individual asset weights limited to $w_i \leq 0.25$.
- Macroeconomic assets:
 - Exposure to Defensive assets not exceeding 50%;
 - Exposure to cyclical assets of at least 20%.

Finally we used as a reference also the Herfindahl Index computed in this way:

$$HI = w \cdot w$$

3.2 Results and Analysis

Table 3: Comparison of Portfolios G and H with the Equally Weighted

Portfolio	DR	Volatility	Sharpe Ratio	Herf. Idx.
Max DR Portfolio	1.257	0.2069	0.2002	0.1996
Max Entropy Portfolio	1.154	0.2050	0.2923	0.0640
Equally Weighted Portfolio	1.1533	0.2096	0.2870	0.0625

Looking at the Herfindhal Index the Diversification Ratio strategy leads to a more concentrated portfolio compared to the Entropy one, showing the outstanding difference caused by the change of definition of Diversification.

4 PCA and Conditional Value-at-Risk

This section develops two portfolios based on different risk-management paradigms: a PCA-based Maximum Sharpe portfolio (Portfolio I) and a CVaR-minimizing allocation (Portfolio J). The objective is to evaluate how statistical factor filtering and tail-risk control shape the resulting risk–return profile.

4.1 Principal Component Analysis (PCA)

We applied PCA to the standardized in-sample covariance matrix to identify the main drivers of systematic risk. Figure 5 shows the variance decomposition, where the first two principal components jointly explain more than 80% of total variance.

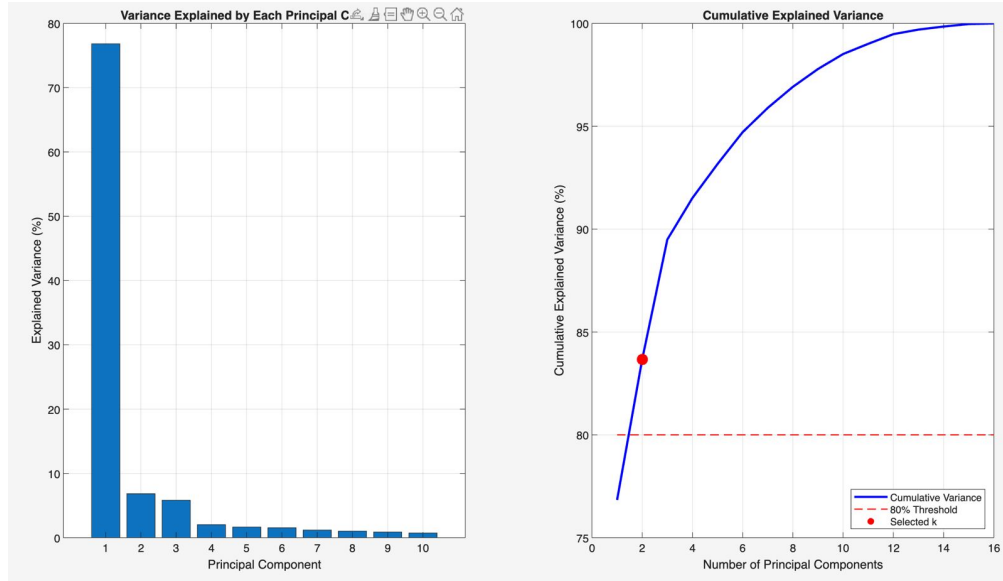


Figure 5: Explained variance by principal components and cumulative explained variance.

4.2 Portfolio I (PCA Maximum Sharpe).

Using the denoised matrix Σ_{PCA} , we computed the maximum Sharpe ratio portfolio subject to standard investment constraints ($\sum_i w_i = 1$, $0 \leq w_i \leq 0.25$) and an annualized volatility cap of 14%. Due to model

underestimation of realized risk (a common characteristic of PCA-denoised matrices), the portfolio achieves an ex-post volatility slightly above the target.

4.3 Portfolio J (Minimum CVaR).

Portfolio J minimizes the 5% Conditional Value-at-Risk (CVaR) of daily returns. The optimization enforces the same investment constraints as above and targets an annualized volatility of 10%. This approach explicitly controls downside tail exposure by penalizing the average loss in the worst 5% of outcomes.

However, the selected target volatility is impossible to achieve with a positive linear combination of the given assets. We therefore tried to minimize the volatility, obtaining a value around 18%.

4.4 Comparative Results

Figure 6 summarizes the differences between the two portfolios: equity curves, drawdowns, return distributions, and risk metrics.

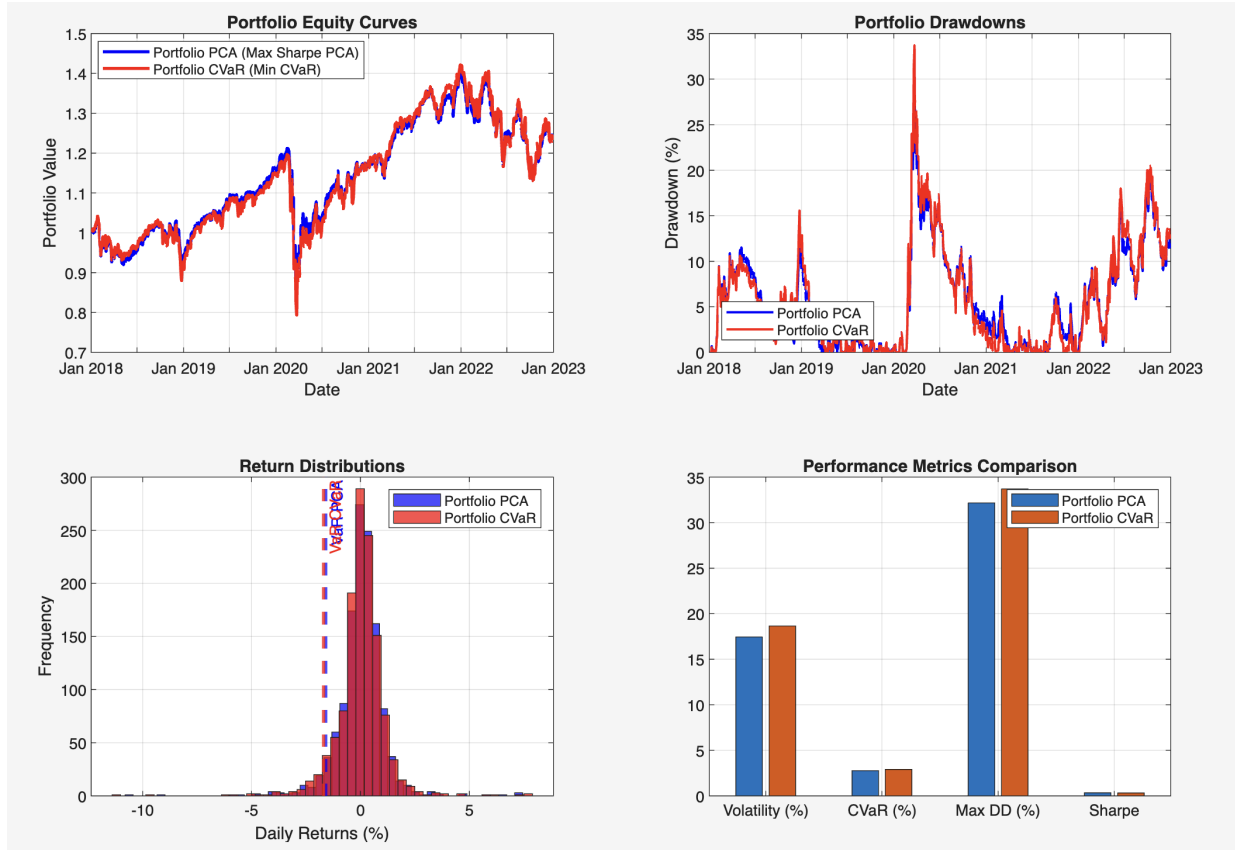


Figure 6: Performance comparison of Portfolio I and Portfolio J.

Below there is the comparison between the two portfolios in terms of Volatility, Cvar, Maximum Draw-down and Sharpe ratio

Table 4: Risk–Return Metrics: PCA vs. CVaR Portfolios

Portfolio	Vol (Ann.)	CVaR (5% Daily)	Max DD	SR
Portfolio I (PCA Max Sharpe)	17.45%	-0.0277	32.16%	0.339
Portfolio J (Min CVaR)	18.66%	-0.0290	33.71%	0.322

5 Exercise 5: Personal Strategy

This section develops a simple allocation rule based on a tiered Sharpe momentum with historical smoothing developed in the in-sample period (2018-2022) and then tested out-of-sample (2023-2024).

5.1 Rationale and Quantitative Intuition

The strategy is based on a monthly ranking of the assets Sharpe ratio over the previous month and the allocation in a fixed portfolio scheme:

- Best 4 : get 15% of the weight of the portfolio each, adding up to 60% of the total.
- Middle 8: get 3.75% of the weight of the portfolio each, adding up to 30% of the total.
- Worst 4: get 2.5% of the weight of the portfolio each, adding up to 10% of the total.

This is then combined with the previous portfolio allocation to perpetuate the historical performance with a linear combination of 75 to 25.

The resulting portfolios are then used to compute the realized monthly returns, volatilities, drawdowns, Sharpe and Calmar ratio over the in-sample time period.

5.2 Performance Evaluation Over Jan 2023 – Nov 2024

As a strategy in the out-of-sample period we use the last portfolio computed in the training period which as previously explained carries over a historical evaluation (we decided to call it **MIX** acronym for **M**onthly **I**ntelligent sharpe **eX**tractor):

$$w_{\text{mix}} = \begin{matrix} 2.83\%, 3.60\%, 2.61\%, 12.14\%, 4.27\%, 3.88\%, 3.51\%, 14.33\%, \\ 5.58\%, 12.71\%, 3.44\%, 5.45\%, 5.86\%, 3.75\%, 12.27\%, 3.77\% \end{matrix}$$

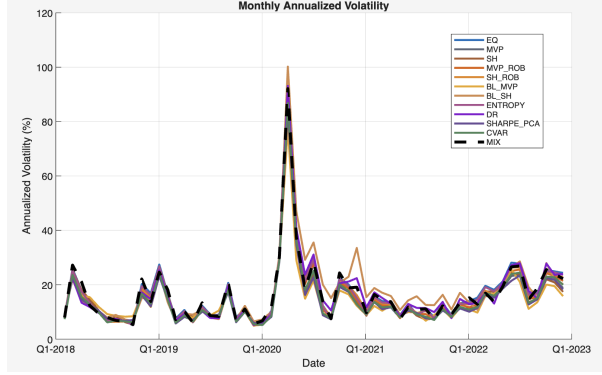
5.3 Discussion of Performance Indicators

In general, the strategy exhibits an intuitive behavior:

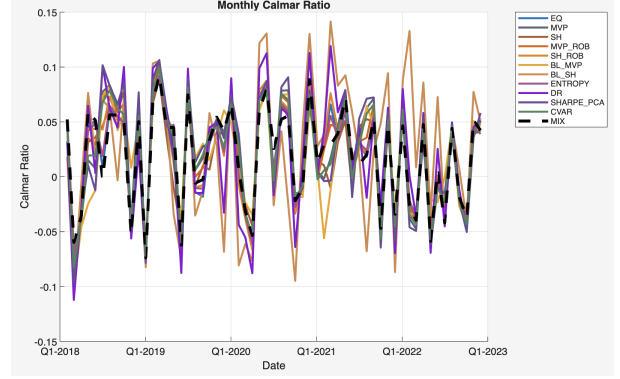
- Returns and Sharpe ratios fluctuate significantly, reflecting changing market conditions across the evaluation period.
- Volatility is generally moderate (generally below 20%), except for a few turbulent months where it spikes.
- Calmar ratios are mainly positive, with some negative spikes predominantly in the 2022-2023 period, which confirm the turbulent time period analyzed.
- The drawdown profile highlights a few severe market events, indicating sensitivity to market conditions and hits a dangerous spike of 27% in Q2 2022, although mitigated compared to all other portfolios which go up to 34%.

In summary, the strategy delivers a balanced combination of moderate returns, stable volatility and occasional drawdown spikes in a very turbulent timeframe, which speak to the goodness of the historical weighted strategy.

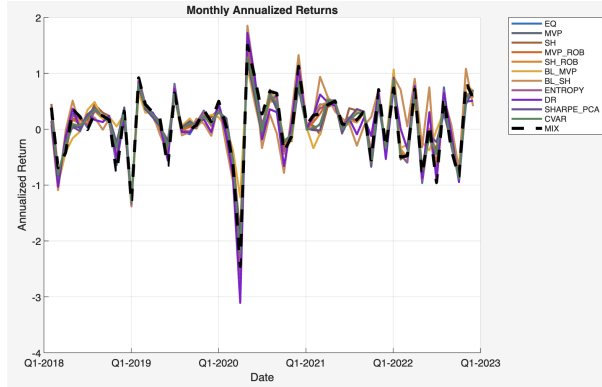
5.4 Monthly Performance Results



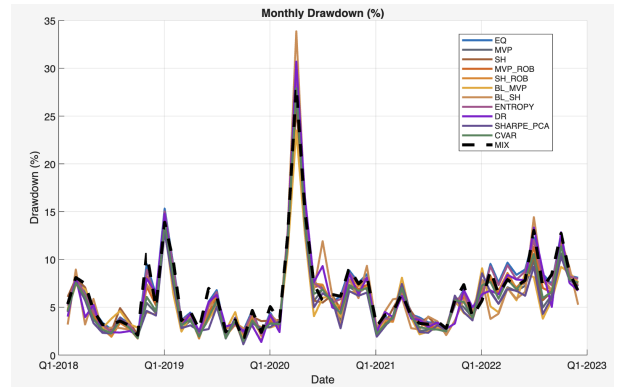
(a) Monthly Annualized Volatility



(b) Monthly Calmar Ratio



(c) Monthly Annualized Returns



(d) Monthly Drawdown (%)

Figure 7: Monthly performance metrics for all portfolios (2018-2023)

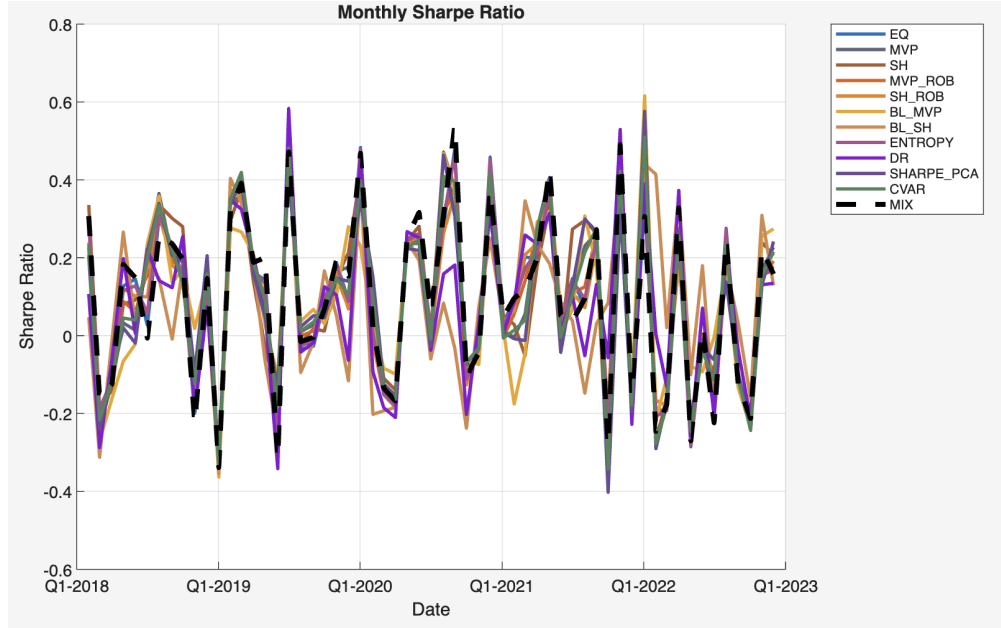


Figure 8: Monthly Sharpe Ratio

5.5 Monthly Comparison of Performance

Table 5: Monthly Performance Summary (In-Sample)

Portfolio	Avg Sharpe	Avg Return	Avg Vol	Avg Drawdown	Avg Calmar
EQ	0.091	0.07	0.17	6.65	0.018
MVP	0.092	0.06	0.14	5.64	0.021
SH	0.097	0.08	0.15	5.96	0.018
MVP_ROB	0.091	0.06	0.15	6.01	0.021
SH_ROB	0.088	0.06	0.16	6.26	0.020
BL_MVP	0.086	0.08	0.14	5.51	0.018
BL_SH	0.072	0.10	0.18	6.16	0.025
ENTROPY	0.092	0.07	0.16	6.49	0.018
DR	0.073	0.05	0.16	6.22	0.021
SHARPE_PCA	0.088	0.06	0.14	5.45	0.022
CVAR	0.093	0.06	0.15	5.81	0.020
MIX	0.095	0.08	0.16	6.54	0.019

6 Weights Summary

This section provides a consolidated overview of the asset allocation weights for all portfolio strategies implemented so far.

Table 6: Unified Asset Allocation Weights Across All Portfolio Strategies (in %)

Asset	EQ	MVP Weights			MSRP Weights			Black–Litterman		DR	Entropy	PCA	CVaR
		Built-in	Nominal	Robust	Built-in	Nominal	Robust	MVP	MaxSharpe				
Asset1	6.2500	0.0000	0.6000	2.5500	30.0000	4.7100	5.0000	0.0000	0.0147	0.0000	5.0600	0.0000	0.5200
Asset2	6.2500	7.6800	8.7000	11.6300	0.0000	6.9700	9.0400	0.3099	0.0000	25.0000	7.5900	10.5400	10.8100
Asset3	6.2500	0.0000	1.2200	5.0700	0.0000	5.5300	4.2000	0.0000	0.0000	13.6100	5.8000	0.0000	3.6600
Asset4	6.2500	30.0000	29.0800	18.9900	11.0000	28.4300	13.9200	0.7340	0.2752	15.8100	8.7700	25.0000	24.9500
Asset5	6.2500	26.2400	20.5900	10.1200	30.0000	28.7800	8.8200	0.2469	0.0000	2.2500	7.3800	25.0000	13.4900
Asset6	6.2500	0.0000	3.9800	3.5200	0.0000	0.4900	6.5800	0.0000	0.0000	7.9800	7.3800	0.0000	1.3000
Asset7	6.2500	11.6700	11.2000	9.3300	0.0000	3.0500	7.6100	0.0000	0.0000	19.1800	6.2700	14.4600	6.1100
Asset8	6.2500	0.0000	1.4400	5.6200	0.0000	0.7300	7.4100	0.0191	0.4002	0.0000	5.7600	0.0000	3.8800
Asset9	6.2500	0.0000	0.0000	0.0000	0.0000	0.0000	4.9100	0.0000	0.0000	23.7100	5.2700	0.0000	0.0000
Asset10	6.2500	0.0000	1.5200	3.4500	0.0000	6.5700	8.5700	0.0000	0.0000	0.0000	5.7800	0.0000	3.4700
Asset11	6.2500	0.0000	2.4600	8.8500	0.0000	0.2000	3.6900	0.0000	0.0000	11.0000	6.2900	0.0000	2.0700
Asset12	6.2500	0.0000	1.1000	0.5500	29.0000	0.2100	0.9700	0.0000	0.0000	0.2000	5.3900	0.0000	1.9000
Asset13	6.2500	13.8300	6.8200	3.4600	0.0000	1.9000	6.5200	0.0000	0.0000	12.0000	6.2500	0.0000	4.3800
Asset14	6.2500	0.0000	1.1400	0.3600	0.0000	0.1200	0.3900	0.0000	0.0000	0.0000	6.0000	0.0000	2.9800
Asset15	6.2500	11.0800	9.6900	5.9600	0.0000	2.4200	3.9100	0.0000	0.0000	0.0000	7.2500	11.1200	10.5500
Asset16	6.2500	0.0000	0.7700	0.5400	0.0000	0.8600	1.2900	0.0000	0.0000	0.0000	5.6000	0.0000	2.4600

7 Final Discussion: Out-of-Sample Analysis (2023–2024)

This section presents a comparative evaluation of all portfolios (A–J) and the portfolio we implemented (MIX) over the out-of-sample dates from January 2023 to November 2024. The analysis is structured around the key performance dimensions: risk metrics, performance metrics and robustness.

7.1 Out-of-Sample Performance of All Portfolios

The out-of-sample period serves as a stringent test for the stability and generalizability of each portfolio construction method. Table 7 summarizes the key performance and risk metrics for all portfolios.

Table 7: Out-of-Sample Performance Metrics (2023–2024)

Portfolio	Ret (%)	Vol (%)	Sharpe	Sortino	Max DD	Calmar	CVaR	Eff N	Div Ratio	Herf
EQ	18.907	11.270	1.678	2.486	10.089	1.874	-0.015	16.000	1.378	0.063
MVP	12.605	9.695	1.300	2.020	10.399	1.212	-0.013	6.154	1.345	0.163
SH	13.726	9.853	1.393	2.232	9.116	1.506	-0.012	4.548	1.334	0.220
MVP_ROB	15.547	10.380	1.498	2.243	10.261	1.515	-0.014	10.525	1.448	0.095
SH_ROB	16.184	10.710	1.511	2.247	10.106	1.601	-0.014	12.791	1.456	0.078
BL_MVP	7.447	9.806	0.759	1.241	11.623	0.641	-0.012	1.667	1.113	0.600
BL_SH	7.571	12.148	0.623	0.961	11.720	0.646	-0.016	3.011	1.368	0.332
ENTROPY	17.997	10.974	1.640	2.442	10.179	1.768	-0.015	15.624	1.384	0.064
DR	16.142	11.314	1.427	2.089	9.501	1.699	-0.016	5.011	1.558	0.200
SHARPE_PCA	11.374	9.660	1.178	1.835	10.090	1.127	-0.013	4.556	1.310	0.220
CVAR	13.683	9.946	1.376	2.117	10.738	1.274	-0.013	8.237	1.348	0.121
MIX	15.904	10.760	1.478	2.254	10.377	1.533	-0.014	11.448	1.333	0.087

7.2 Risk–Return Analysis

Using Table 7, portfolios are classified according to out-of-sample performance metrics mainly returns, volatility and drawdowns.

1. **Low Risk / Low Return.** Portfolios **MVP**, **BL_MVP**, **SHARPE_PCA** and **SH** fall in this quadrant. They exhibit subdued volatility and relatively mild drawdowns, but also deliver the lowest average returns in the universe. These allocations are suited to very conservative investors who mainly seek stability and capital preservation.

2. **Low Risk / High Return.** Portfolios **CVAR**, **MVP_ROB** and **SH_ROB** achieve returns comparable to the best performers while keeping volatility around the 10% threshold. This results in attractive Sharpe ratios and efficient risk-taking. Both portfolios combine strong performance with good diversification and controlled tail risk, making them an appealing “efficient growth” choices.
3. **High Risk / Low Return.** **BL_SH** is the only portfolio clearly in this category. It carries volatility above 10% but does not compensate investors with sufficiently high returns, leading to weaker risk-adjusted metrics. This configuration is the least desirable, as investors take more risk without a corresponding improvement in performance.
4. **High Risk / High Return.** Portfolios **EQ**, **DR**, **ENTROPY** and **MIX** all display both elevated volatility and high average returns. They sit in the upper-right region of the risk–return space, with strong Sharpe and Sortino ratios but also the deepest maximum drawdowns. These allocations are appropriate for investors with a high risk appetite who are willing to accept substantial fluctuations in exchange for superior expected growth.

This quadrant-based classification highlights the trade-offs across portfolios: moving from the low-risk/low-return cluster to the high-risk/high-return one increases both performance potential and drawdown exposure, while the low-risk/high-return group provides the most attractive compromise between return and risk.

The great performance of the equally weighted portfolio is to be expected out-of-sample as noted in the paper: Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy? by Victor DeMiguel, Lorenzo Garlappi and Raman Uppal in 2009 (available in the zip archive). Their work demonstrates that when asset returns are difficult to forecast, a common reality in practice, this "naive" diversification strategy is a solid and often optimal strategy.

Robustness Across Sub-Periods

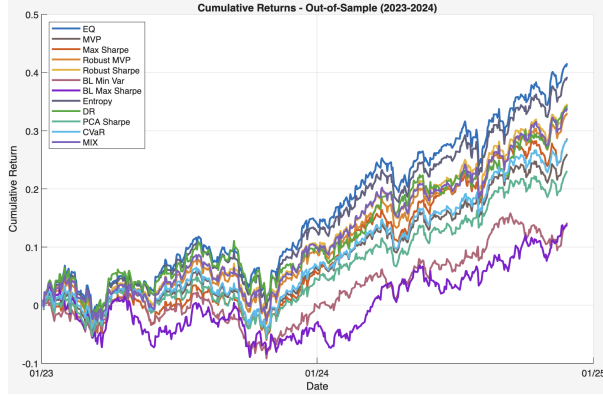
Table 8 reports the stability of Sharpe ratios and volatilities across four sub-periods of the out-of-sample window.

Table 8: Robustness Metrics: Stability of Performance Across Time

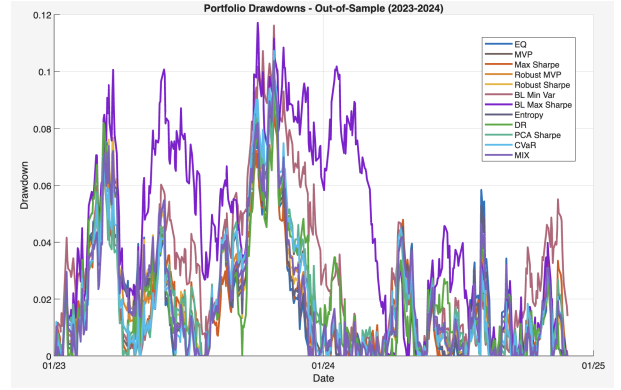
Portfolio	Sharpe Stability	Vol Stability
Portfolio_EQ	0.3959	0.1411
Portfolio_MVP	0.7999	0.1401
Portfolio_SH	0.9274	0.0864
Portfolio_MVP_ROB	0.6352	0.1485
Portfolio_SH_ROB	0.5584	0.1566
Portfolio_BL_MVP	1.5462	0.0800
Portfolio_BL_SH	1.6488	0.1605
Portfolio_ENTROPY	0.4499	0.1411
Portfolio_DR	0.6295	0.1523
Portfolio_SHARPE_PCA	0.9709	0.1407
Portfolio_CVAR	0.7511	0.1397
Portfolio_MIX	0.5437	0.1585

Lower values indicate greater stability. The Black–Litterman portfolios show high Sharpe instability, reflecting sensitivity to view misspecification. Entropy, MIX and Robust portfolios exhibit the most stable Sharpe performance, meanwhile SH and BL_MVP exhibit the most stable Vol performance.

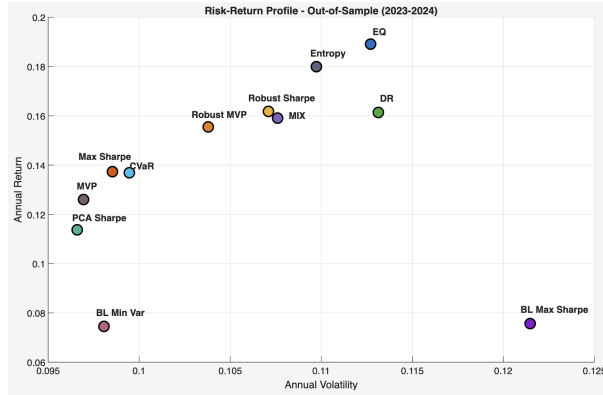
7.3 Out-of-Sample Performance Visualizations



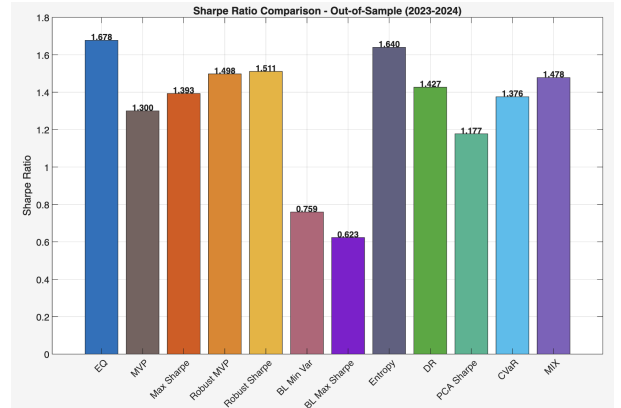
(a) Cumulative Returns (2023-2024)



(b) Risk-Return Profile (2023-2024)



(c) Portfolio Drawdowns (2023-2024)



(d) Sharpe Ratio Comparison (2023-2024)

Figure 9: Out-of-sample performance analysis of all portfolios (2023-2024)

7.4 Impact of macro-sensitivities on Portfolio behaviour

This section examines how macro-sector exposures influence the out-of-sample risk-return characteristics of the portfolios.

Global Relationships

The scatter plots in Figure 10 and the correlation analysis reveal two dominant patterns:

- Cyclical exposure is **strongly negatively correlated** with Sharpe ratio ($\rho_{\text{cyc}, \text{Sh}} = -0.802$) and **positively correlated** with maximum drawdown ($\rho_{\text{cyc}, \text{MaxDD}} = 0.757$).
- Defensive exposure shows the opposite behavior: it is **positively correlated** with Sharpe ($\rho_{\text{def}, \text{Sh}} = 0.196$) and **negatively correlated** with maximum drawdown ($\rho_{\text{def}, \text{MaxDD}} = -0.417$).

These patterns indicate that increasing Cyclical weight generally worsens risk-adjusted performance, while increasing Defensive weight stabilizes returns and reduces tail losses.

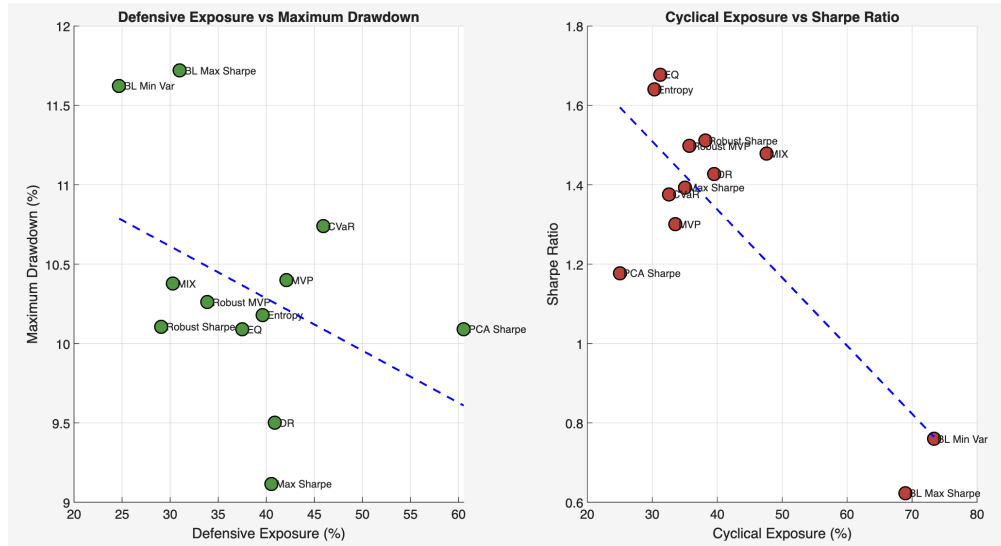


Figure 10: Defensive Exposure vs. Maximum Drawdown and Cyclical Exposure vs. Sharpe Ratio.

Portfolios with substantial Defensive allocations (PCA Sharpe, CVaR, MVP)—often exceeding 40%—demonstrate low volatility and solid Sharpe ratios (in the 1.2-1.4 range), meanwhile Black-Litterman portfolios, where we allocate more than 70% to Cyclical sectors, record the lowest Sharpe ratios and the deepest maximum draw-downs (above 11.5%).

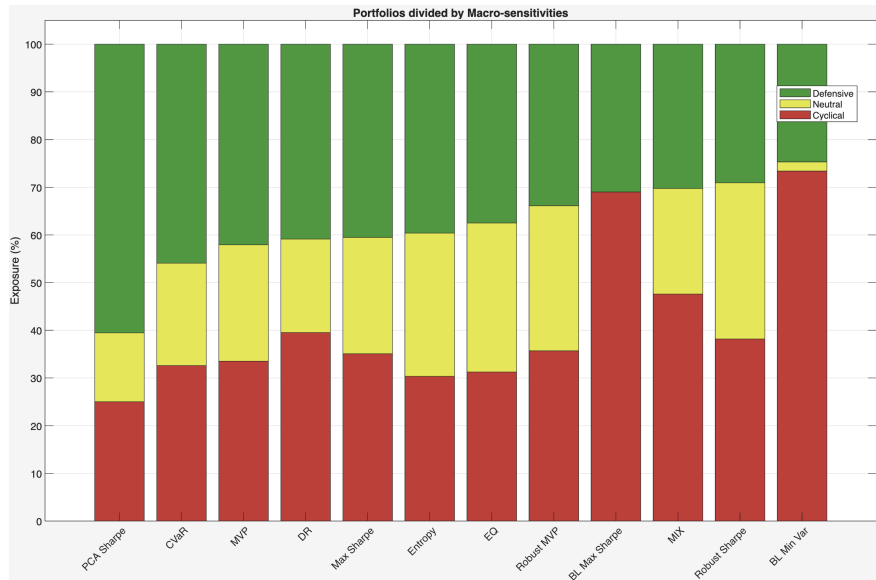


Figure 11: Macro-sector exposure decomposition across portfolios.

Overall, the data suggest that portfolios with controlled Defensive exposure and balanced macro-structure produce more stable and efficient performance than those taking large one-sided cyclical bets.